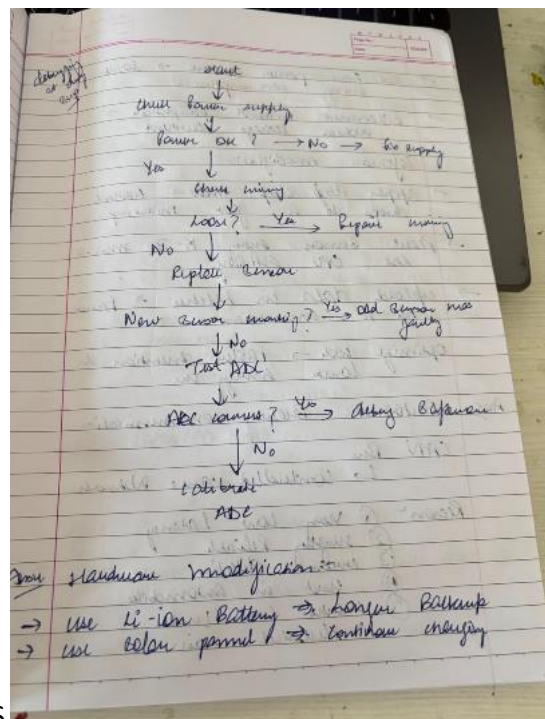
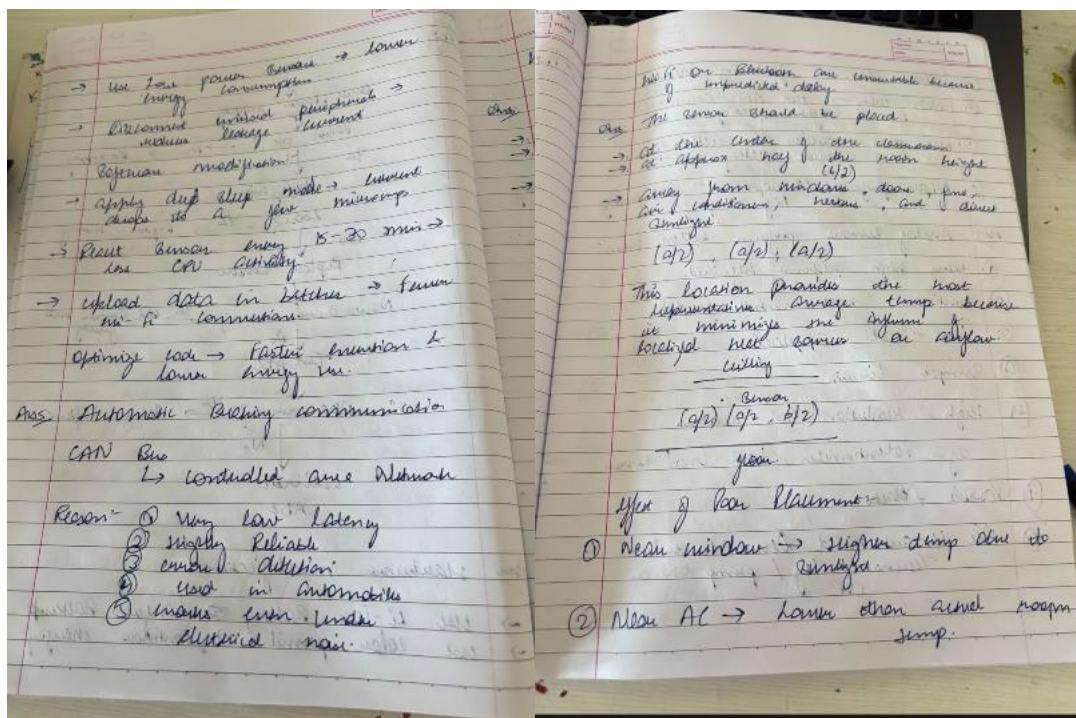
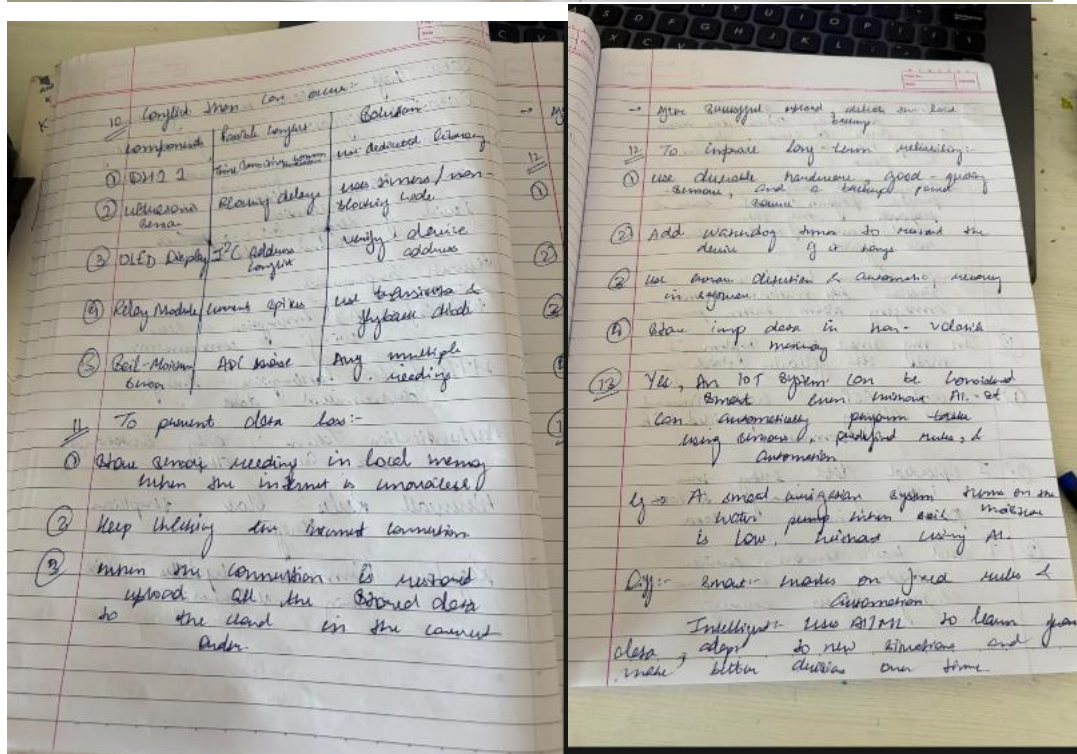
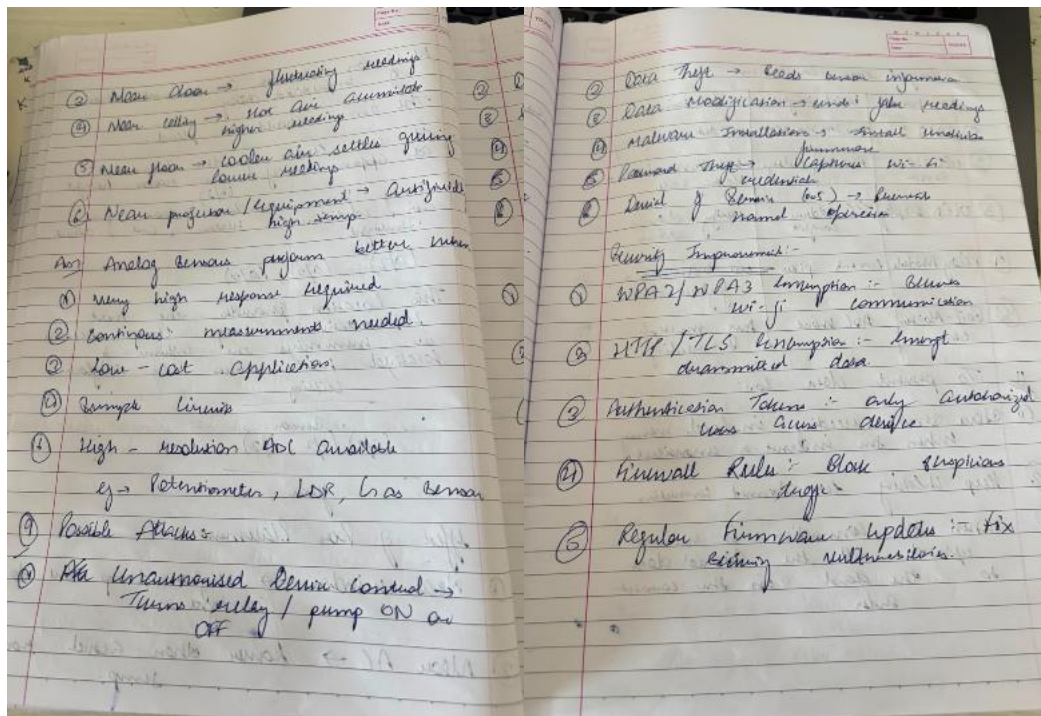


NAME- YUVIA



BATCH- IOT & DRONES





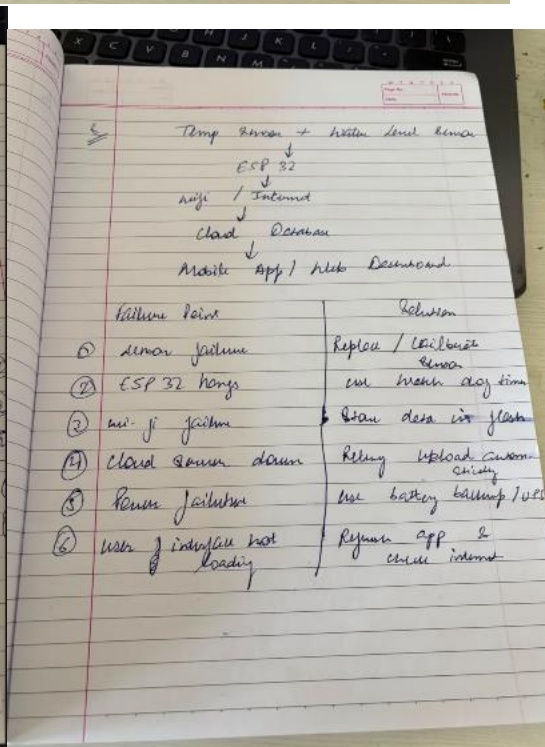
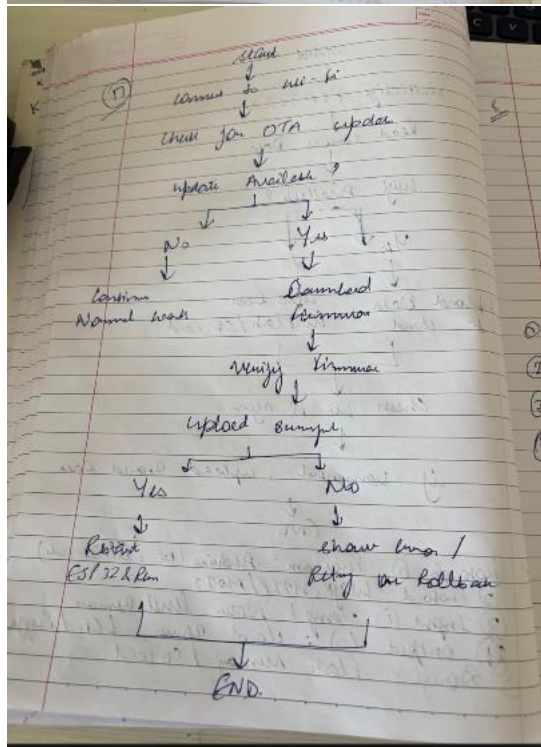
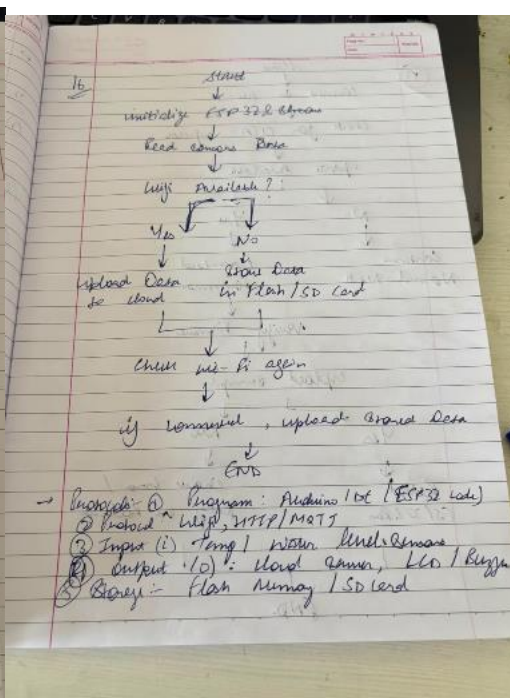
16. Rolling means the processor continuously checks the status of device at given interval. It is simple but needs CPU, some power.

Intelligence based programming means the processor performs other tasks & responds when event occurs. It is faster & more efficient.

Advantage of IoT programming is function like motion detection, emergency alarm, button press.

17. In my smart water system, I used the following code.

1. I used ESP 32, which I received with slightly low provided WiFi connectivity.
2. I uploaded data every min instead of continuously to save power & instead data.
3. I used basic sensors, which needed less but gave slightly lower accuracy.



- it can easily need an external module sensor
- it can control relay
- cause cost and cause power consumption
- the program is simple and requires little memory

Features	Wi-Fi	Bluetooth	LoRa	MQTT
Range	10-100m	10-100m	2-15km	depends on network
Rate	very high	medium	low	medium
Power consumption	High	Low	Very low	High
Internet support	Yes	No	Yes	Yes
No. of devices	unlimited	limited	Thousands	Thousands
Bit rate	High-speed data	medium	Long-range	Cloud communication

For 500 sensors distributed across a 2km area, the protocol or combination of protocols that can be chosen are LoRa and MQTT.

Benefits of LoRa:

- Covers entire 2km factory
- Supports thousands of sensors
- Very low battery consumption
- Reliable in industrial environment.

Benefits of MQTT

- Lightweight communication protocol
- Efficient cloud communication
- Supports Publish-Subscribe Architecture
- Reliable even with unstable networks

The combination of take for long range sensor communication and MQTT for cloud communication provides the most scalable, reliable, and energy-efficient solution.

Ans To determine whether the problem lies in the sensor, wiring, power supply, ADC or software, always isolate one component at a time instead of changing everything randomly.

eg. If it is a faulty sensor, it will show inconsistent readings constantly. If it shows many analog values, there is an ADC problem.

Ques You are building a smart irrigation system was not required

Ans I would choose the ESP 32 because it has built-in wi-fi & bluetooth, making it easy to send sensor data to a mobile application every 5 sec. it is also much faster than the Arduino Uno, has more memory, and supports multitasking.

To monitor soil moisture, control a water pump, and send data to a mobile application every 5 sec, we need ESP32 because: it has built-in wi-fi which enables direct communication with the mobile application. faster processor allows simultaneous sensor reading, relay control, and cloud communication.

Bluetooth can be used for future expansion.

More RAM & flash support larger programs.

if internet is not required, then Arduino Uno is a suitable and economic option because: